

For 05 finne maksimumspenget
Ser man etter hvor den deriverte
av $f(x)$ er lik 0,
Så hvor $f'(x) = 0$

$$f(x) = e^{-x/4} \cdot \tan^{-1}(x)$$

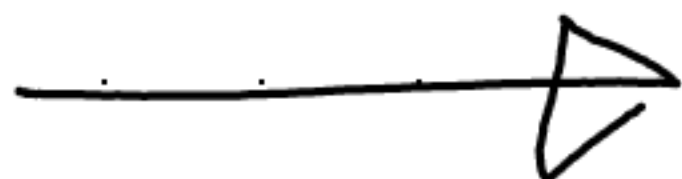
$$f'(x) = u' \cdot v + u \cdot v'$$

$$u = \underline{e^{(-\frac{1}{4})x}}, \quad u' = ke^{kx} = \underline{-\frac{1}{4}e^{(-\frac{1}{4})x}}$$

$$v = \underline{\tan^{-1}(x)}, \quad v' = \underline{\frac{1}{1+x^2}}$$

$$f'(x) = -\frac{1}{4}e^{(-\frac{1}{4})x} \cdot \tan^{-1}(x) + e^{(-\frac{1}{4})x} \cdot \frac{1}{1+x^2}$$

$$f'(x) = e^{(-\frac{1}{4})x} \left(-\frac{1}{4}\tan^{-1}(x) + \frac{1}{1+x^2} \right)$$



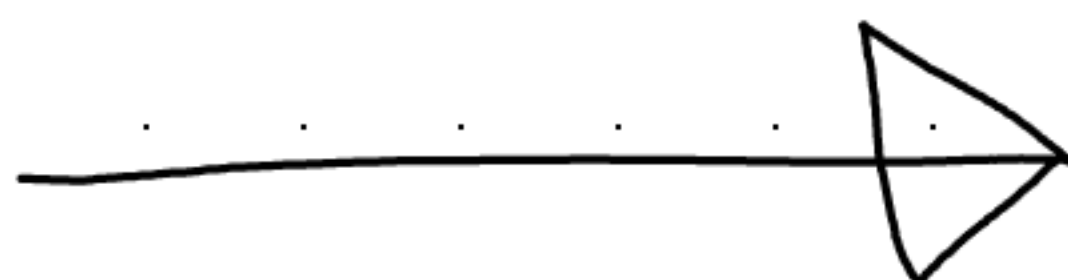
$$f'(x) = e^{(-\frac{1}{4})x} \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right)$$

Sjekker hvor dette er lik 0

$$\begin{array}{l|l} e^{(-\frac{1}{4})x} \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right) = 0 & \div e^{(-\frac{1}{4})x} \\ -\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} = 0 & \cdot 4 \\ -\tan^{-1}(x) + \frac{4}{1+x^2} = 0 & \cdot (-1) \\ \underline{\underline{\tan^{-1}(x) - \frac{4}{x^2+1} = 0}} & \end{array}$$

Så maksimal punkt i
funksjonen bestemmes med
dette uttrykket.

(Som er identisk til den i
obcis teksten)



Jes skul bruke Newtons metode for den numeriske
delen, så jeg får å finne den andre deriverte

$$f'(x) = e^{(-\frac{1}{4})x} \cdot \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right)$$

$$f''(x) = u' \cdot v + u \cdot v'$$

$$u = e^{-x/4}, \quad u' = -\frac{1}{4} e^{-x/4}$$

$$v = \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right)$$

$$v' = \left(-\frac{1}{4} \cdot \frac{1}{1+x^2} - \frac{2x}{(1+x^2)^2} \right) = \left(-\frac{1}{4(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$

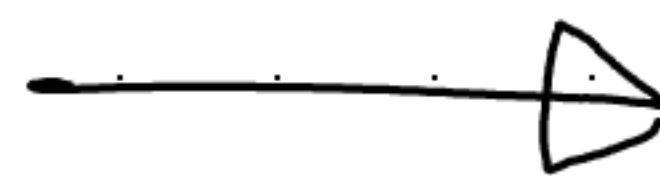
$$= -\frac{1}{4} e^{-x/4} \cdot \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right) + e^{-x/4} \cdot \left(-\frac{1}{4(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$

$$= e^{-x/4} \left(-\frac{1}{4} \left(-\frac{1}{4} \tan^{-1}(x) + \frac{1}{1+x^2} \right) - \frac{1}{4(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$

$$= e^{-x/4} \left(\frac{1}{16} \tan^{-1}(x) - \frac{1}{4(1+x^2)} - \frac{1}{4(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$

$$= e^{-x/4} \left(\frac{1}{16} \tan^{-1}(x) - \frac{2}{4(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$

$$f''(x) = e^{-x/4} \left(\frac{1}{16} \tan^{-1}(x) - \frac{1}{2(1+x^2)} - \frac{2x}{(1+x^2)^2} \right)$$



Jes Skal da bruge:

$$X_{n+1} = X_n - \frac{f'(X_n)}{f''(X_n)}$$

i den numeriske del
for og finde x verdier
for maksimale punkt.

Jes lige at maksimale punkter
ligger i:

$$\underline{\underline{X_m \approx 1,69070816}}$$

og

$$\underline{\underline{f(X_m) \approx 0,67932226}}$$